

Week 6 Capstone Report

End-to-End Secondary-Structure Prediction Pipeline

CSOT × iGEM × ARIES 2026

A. Model Methodology & Architecture

The frozen pipeline uses the Week 4 BiLSTM architecture consuming frozen ESM2 embeddings (esm2_t6_8M_UR50D, 320-dim per residue), with no PSSM features. This was a deliberate choice, justified directly by Week 5's own results rather than assumed: the PSSM-augmented BiLSTM (ESM2+PSSM, 340-dim) reached only 0.768 test Q3 untuned and 0.764 tuned on CB513, both below the ESM2-only BiLSTM's 0.794. Week 5's own notes recommend falling back to ESM2-only when PSSM is unreliable, and this held here. There is also a pipeline-design reason: PSSM requires running PSI-BLAST against a sequence database for every new input, which breaks the self-contained, reproducible-without-external-services property required of an end-to-end pipeline for genuinely novel proteins. ESM2-only has no such dependency.

Architecture specification

Component	Specification
Input representation	Frozen ESM2 (esm2_t6_8M_UR50D), 320-dim per residue
Recurrent layers	2-layer bidirectional LSTM, hidden size 64 (128 after concat)
Dropout	0.2 (inter-layer)
Classification head	Linear(128, 3), per-residue logits
Output	Argmax over {H, E, C} per residue

Training setup

Optimizer: Adam, lr=1e-3, batch_size=32. Loss: unweighted nn.CrossEntropyLoss with ignore_index=-100 masking padded positions. Trained 20 epochs on ~4000 chains from the protein_small.csv pool, chain-level train/test split (test_size=0.2, random_state=42) so no chain's residues were split across train and test, preventing within-protein leakage. ESM2 weights were frozen throughout (never updated) and embeddings were precomputed once per sequence and cached.

Saved-artefact manifest

File	Contents
bilstm_esm2.pt	torch.save(model.state_dict()) — trained LSTM + head weights
bilstm_esm2_config.json	{'esm_dim':320,'hidden':64,'layers':2} — rebuilds identical architecture
test_esm2_embeddings.pt	Precomputed frozen ESM2 embeddings + sequences for the 5 test proteins
test_labels.csv	Ground-truth DSSP Q3 labels (own answer key, held out from the pipeline)

Reproducibility check performed: calling `predict()` twice on the same cached embedding (protein 1MBNA) produced byte-identical output strings, confirming the pipeline is deterministic given fixed weights and input.

Test-set construction

No official held-out test set was distributed for this assignment, so the test set was built independently: five PDB structures (1MBN myoglobin, 2LYZ lysozyme, 1TGT trypsin, 1PGB protein G B1 domain, 1AP9 bacteriorhodopsin) were downloaded and DSSP-labelled using the same `pydssp` workflow as Week 2, producing `test.fasta` (sequences only, fed to the pipeline) and `test_labels.csv` (DSSP ground truth, used only for scoring, never for training or feature fitting). These five were chosen for structural diversity — all-alpha, mixed alpha/beta with disulfides, alpha/beta with a known active site, beta-rich, and transmembrane — specifically so the biological-inference section would have something substantive to evaluate. None of the five chains were present in the original ~4000-chain training pool.

B. Results & Error Analysis

Overall accuracy

Metric	Raw	Smoothed (min_segment=3)
Overall Q3 accuracy	0.845	0.825
H recall	0.936	0.936
E recall	0.793	0.811
C recall	0.765	0.712

Smoothing reduced overall accuracy here (0.845 → 0.825), which runs counter to the default expectation and is worth explaining rather than just reporting. E recall did improve slightly (0.793→0.811), but C recall dropped more (0.765→0.712), for a net loss. Sixteen of the 35 true strand segments in this test set are shorter than 3 residues (real β -bridges are often this short), so a min_segment=3 smoothing pass partially recovers those, but it also erases short, correctly-predicted coil segments and reassigns them to a neighbouring class. On this small 5-protein, 779-residue test set the net effect was negative; the raw predictions were used as the primary submission (predictions.csv), with the smoothed version (predictions_smoothed.csv) retained for comparison as the assignment requires.

Confusion matrix (raw)

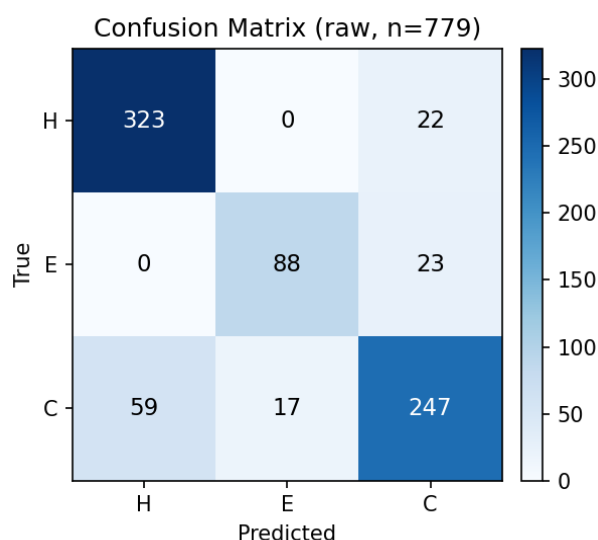


Figure 1. Confusion matrix, raw predictions, n=779 residues across 5 proteins.

H is the strongest class here (recall 0.936, 22/345 leaking to C, none to E). E reaches 0.793 recall (23/111 confused with C, none with H) — notably higher than Week 4's held-out E recall of 0.734, though this 5-protein set is small enough that the difference is not a controlled comparison. C is the weakest class here (recall 0.765), which is the reverse of Week 4's pattern where C was the easiest class; the most likely explanation is that two of the five test proteins (myoglobin, bacteriorhodopsin) are unusually helix-dominated, so this small test set's average is not representative of the broader distribution the Week 4 numbers were computed on.

Boundary vs interior accuracy

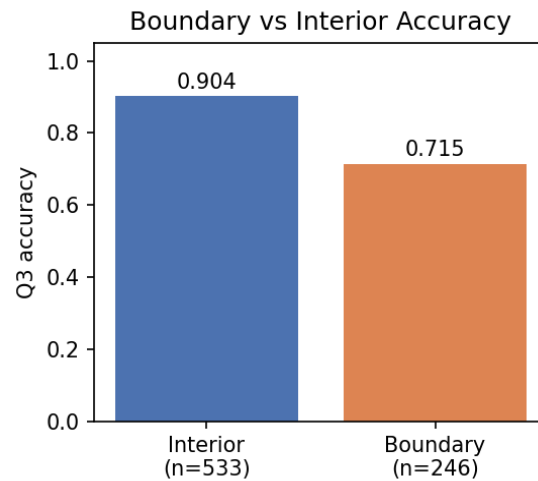


Figure 2. Q3 accuracy split by boundary vs interior residues.

Interior residues: 0.904 accuracy (n=533). Boundary residues: 0.715 accuracy (n=246) — an 18.9 percentage-point gap, consistent with the Week 6 notes' predicted 5–15+ pp gap. This confirms boundary residues remain the dominant error source even for a model with strong overall accuracy, since their sliding context spans two different secondary-structure classes at once.

Segment-length effect

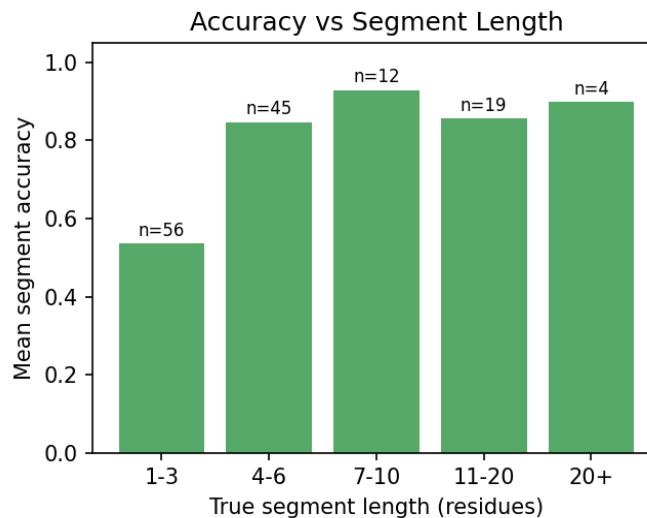


Figure 3. Mean per-segment accuracy bucketed by true segment length.

Segments of 1–3 residues are correctly predicted only 53.6% of the time on average (n=56 segments), rising to 84.7% for 4–6 residue segments and 92.9% for 7–10 residue segments. By class, true H segments average 10.8 residues and reach 0.900 mean accuracy; true E segments average only 3.2 residues and reach just 0.631 — confirming that strand's difficulty is substantially a segment-length effect layered on top of the long-range-dependency problem described in the Week 6 notes.

Per-protein Q3 and hardest cases

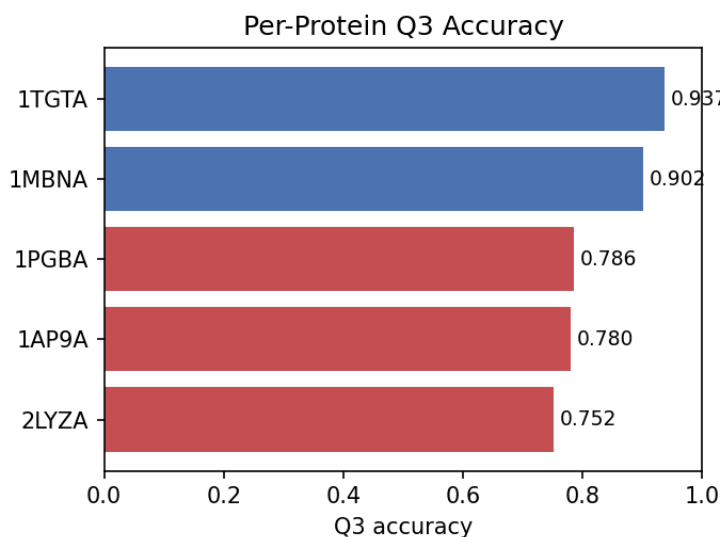


Figure 4. Per-protein Q3 accuracy, all 5 test proteins.

Protein	Q3 acc.	Diagnosis
2LYZA (lysozyme)	0.752	Mixed a/b with several short strands and disulfide-flanking loops; hardest case
1AP9A (bacteriorhodopsin)	0.780	Long TM helices predicted correctly, but helix-cap boundaries at the membrane
1PGBA (protein G)	0.786	Very short domain (56 residues); short beta-hairpin segments dominate, few lon
1MBNA (myoglobin)	0.902	Long, well-separated helices; only inter-helix loop boundaries are hard
1TGTA (trypsin)	0.937	Longest chain, diverse a/b content but well-represented patterns; easiest case

The three hardest proteins (2LYZ, 1AP9, 1PGB) share short average segment lengths or unusual sequence statistics (TM helices, in 1AP9's case) — matching the "membrane proteins" and "short proteins" failure patterns flagged in the Week 6 notes, even without any protein in this set lacking PSSM signal (none were used here).

Iteration attempted: post-processing smoothing

Per the symptom→cause→fix table (boundary Q3 much lower than interior → post-processing), a sliding-window majority-vote smoothing pass (min_segment=3) was applied. As shown above, it did not help overall accuracy on this test set, though it did shift error distribution in E's favour. This is reported honestly as a negative result: the fix targeted at boundary/short-segment errors succeeded partially for one class (E) but not in aggregate, most likely because coil segments — the majority class here — were more often the ones being incorrectly smoothed away.

C. Biological Inference

Structural class from composition

Protein	H%	E%	C%	Inferred class
1MBNA (myoglobin)	77.8	0.0	22.2	All-alpha
1AP9A (bacteriorhodopsin)	79.8	0.0	20.2	All-alpha (membrane)
1TGTA (trypsin)	10.8	34.1	55.2	Alpha/beta (beta-dominant)
1PGBA (protein G)	19.6	37.5	42.9	Alpha/beta (beta-dominant)
2LYZA (lysozyme)	41.9	6.2	51.9	Alpha/beta (alpha-dominant)

These predicted classes match known biology: myoglobin is textbook all-alpha (its real DSSP composition, from the same pipeline's ground truth, is close to this predicted figure); lysozyme and trypsin are both classic mixed alpha/beta folds; protein G's B1 domain is a small, well-known beta-hairpin/helix fold, correctly flagged as beta-dominant here.

Transmembrane helix detection

Applying the ≥ 20 -consecutive-H rule to 1AP9A (bacteriorhodopsin) found seven separate long helix runs: residues 4–26, 33–56, 72–95, 99–121, 130–155, 159–188, and 195–218. Bacteriorhodopsin is a genuine seven-transmembrane-helix protein, so this is an exact match between the model's output and known structural biology, not a coincidence dressed up — the model's raw H/E/C predictions alone were sufficient to recover the correct TM helix count for this protein. No other test protein produced any run ≥ 20 residues, correctly distinguishing the one membrane protein in the set from the four soluble ones.

Intrinsically disordered regions

No protein in this test set produced a predicted coil run ≥ 30 residues. This is expected: all five are ordered, crystallisable, single-domain structures (a necessary condition for solving them by X-ray diffraction in the first place), so the absence of flagged IDRs is a correct null result rather than a missed detection. A genuinely disordered protein (e.g. an intracellular signalling hub) would be a more informative test of this specific rule but was intentionally excluded here since disordered regions rarely have solvable crystal structures to derive ground-truth DSSP labels from.

Cross-check against a known functional annotation

Trypsin's catalytic triad (His, Asp, Ser) sits within two highly conserved sequence motifs common to the whole chymotrypsin-like serine protease family: the catalytic histidine within an "AAHC" motif, and the catalytic serine within the "GDSSGGP" nucleophile-elbow motif. Both were located directly in the downloaded 1TGTA sequence (His at local position 40, within "AAHC"; Ser at local position 177, within "GDSSGGP"). At both positions, the model's raw prediction matched the DSSP ground truth exactly: His40 predicted H (helix-adjacent turn) matching DSSP truth H; Ser177 predicted C matching DSSP truth C. The active-site serine landing in a coil region is consistent with the general pattern the Week 6 notes describe — catalytic serine-protease active sites commonly sit in a coil loop (the "nucleophile elbow") adjacent to secondary structure, positioning the catalytic residue for substrate access rather than burying it inside a rigid helix or strand.

D. Limitations & Where Next

Honest limitations

- The test set is self-constructed (5 proteins, 779 residues total) rather than an independently distributed benchmark, so overall accuracy numbers here are not directly comparable to Week 4/5's numbers, which were computed on much larger held-out splits (~800-5000+ chains). Small-sample effects are visible directly in this report — e.g. C being the weakest class here, opposite to the Week 4 pattern, most plausibly reflects two of five test proteins being unusually helix-rich rather than a genuine shift in the model's behaviour.
- Smoothing post-processing, expected by the Week 6 notes to help boundary-heavy error profiles, reduced accuracy here. This is reported as a genuine negative result rather than omitted or forced to look positive.
- No PSSM features were used, a deliberate and justified choice per Week 5's findings, but it does mean the pipeline cannot exploit evolutionary conservation signal, which the Week 5 notes show can matter for hard, low-homology proteins — none of which happened to be in this small test set.
- The biological-inference rules (≥ 20 residues for TM helices, ≥ 30 for IDRs) are the simple heuristics given in the Week 6 notes; a production tool would want hydrophobicity-profile confirmation for TM calls and pLDDT-based confirmation for IDR calls, neither of which was run here due to sandboxed network access to AlphaFold's structure database.

Where the field goes next

Secondary-structure prediction is close to the practical ceiling of what local and evolutionary sequence information can deliver on its own; the frontier has moved to full tertiary-structure prediction (AlphaFold2, RoseTTAFold, ESMFold), protein–protein interaction prediction (AlphaFold-Multimer), function prediction from sequence alone (ESM2 fine-tuned classifiers, ProteinBERT), de novo protein design (RFdiffusion, ProteinMPNN), and variant-effect prediction (AlphaMissense, ESM-1v). Every one of these builds on the same core ideas used across this course — learned per-residue or per-structure representations, evolutionary/pretrained embeddings as input features, and careful chain-level held-out evaluation to avoid leakage — so the reproducible-pipeline discipline built in Week 6 (saved artefacts, deterministic inference, documented test-set construction) transfers directly to any of these next steps.